Knowledge Cutoff Effects on LLM-Generated Literature Reviews in Rapidly Evolving Fields

Abstract

Large language models (LLMs) are increasingly being used to search, summarize, compare, and synthesize scientific literature. Their usefulness for literature review, however, is constrained by the temporal boundaries of their training data. A conventional knowledge cutoff creates a mismatch between a model's internal parametric knowledge and the continuously changing scientific record, particularly in rapidly evolving fields such as artificial intelligence, biotechnology, medicine, climate science, and computational science. This paper examines how knowledge cutoffs affect LLM-generated literature reviews, distinguishing the reported cutoff of a model from its effective temporal knowledge boundary. The review synthesizes research on parametric knowledge, temporal generalization, hallucination, citation reliability, retrieval-augmented generation (RAG), knowledge editing, temporal knowledge graphs, and automated survey-generation systems. Evidence indicates that cutoff effects extend beyond simple omission of recent papers: they can produce outdated claims, distorted research landscapes, temporal inconsistencies, incomplete citation coverage, and misleading synthesis in which older evidence is presented as representative of the current state of a field. Recent systems such as RAG, Self-RAG, OpenScholar, and time-aware retrieval frameworks substantially mitigate these problems by introducing external and updateable evidence. Nevertheless, retrieval does not automatically solve temporal reasoning, source selection, citation attribution, or synthesis quality. The paper argues that future literature-review systems should treat temporal provenance as a first-class property of scientific evidence and evaluate outputs using freshness, temporal consistency, citation accuracy, coverage, and longitudinal robustness rather than conventional factuality metrics alone. A research agenda is proposed for dynamic scholarly benchmarks, time-aware retrieval, evidence graphs, cutoff-aware evaluation, and continuously auditable literature-review agents.

Keywords: large language models; knowledge cutoff; literature review; scientific literature synthesis; temporal knowledge; retrieval-augmented generation; hallucination; citation accuracy; temporal reasoning; research agents

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1. Introduction

Literature review is a foundational activity in scientific research. Researchers must identify relevant publications, determine how findings relate to one another, distinguish established knowledge from emerging evidence, and formulate conclusions that accurately represent the state of a field. The task has become increasingly difficult because scientific publication is expanding rapidly while research questions themselves are becoming more interdisciplinary and specialized. Recent work on automated scientific synthesis explicitly identifies the growth of the literature as a major obstacle to maintaining comprehensive and current knowledge (Asai et al., 2026).

Large language models have created a new possibility: instead of manually searching hundreds of papers, researchers can ask an LLM to summarize a topic, identify influential approaches, compare methods, or generate a preliminary survey. Early studies established that pretrained language models contain substantial factual and relational knowledge within their parameters (Petroni et al., 2019). However, parametric knowledge is fundamentally tied to the data and optimization process used to train the model. Consequently, an LLM that is not connected to an external information source cannot automatically incorporate scientific discoveries, revised results, newly published papers, or changes in scientific consensus occurring after its effective training period.

This temporal limitation is commonly described through the concept of a **knowledge cutoff**. The conventional interpretation is that information beyond a specified date is not contained in the model's training data. Recent research shows that this definition is too simplistic. Cheng et al. (2024) demonstrate that an LLM may have different *effective cutoffs* for different resources and topics, and that the effective cutoff can differ substantially from a provider-reported cutoff because of temporal biases and data deduplication.

The implications are especially serious for literature reviews. A literature review is not merely a collection of factual statements; it is a temporally situated representation of a research field. If a model stops "seeing" the literature at time (t_c), while the field continues to develop until time (t_q), the resulting review may be systematically incomplete. The problem can be represented as:

$$
\Delta t = t_q - t_c
$$

where (t_c) is the effective knowledge boundary of the model and (t_q) is the temporal boundary relevant to the research question. As (Δt) increases, the probability of missing relevant developments generally increases, although the relationship is neither linear nor uniform across disciplines.

The central argument of this paper is therefore that **knowledge cutoff effects should be understood as a scientific synthesis problem rather than simply an information-retrieval problem**. A model can produce grammatically excellent and apparently coherent text while describing an obsolete research landscape. Moreover, even when recent documents are supplied through retrieval, the model must determine which evidence is temporally relevant, reconcile conflicting findings, preserve historical distinctions, and correctly attribute claims.

This paper reviews the relevant literature and examines five interconnected questions:

1. What exactly constitutes a knowledge cutoff in an LLM?
2. How does a cutoff affect literature-review completeness and scientific synthesis?
3. Why do retrieval-based systems reduce but not eliminate temporal failures?
4. What approaches currently address knowledge freshness and citation reliability?
5. What research gaps must be addressed before LLM-generated literature reviews can be considered reliable research infrastructure?

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2. Background and Related Work

2.1 Parametric Knowledge in Language Models

Pretrained language models acquire information from large textual corpora during training. Petroni et al. (2019) demonstrated that pretrained models such as BERT can recover substantial relational knowledge directly from their parameters. This work established an important conceptual foundation: language models can function partly as implicit knowledge stores even without an explicit database.

The advantage of parametric knowledge is efficiency. Once learned, information can be accessed without an external retrieval operation. The disadvantage is that updating knowledge generally requires additional training, adaptation, editing, or an external source. Cao et al. (2024) conceptualize this problem as a knowledge life cycle involving knowledge acquisition, representation, maintenance, updating, and use. Their survey emphasizes that knowledge in pretrained language models should be viewed as a dynamic lifecycle rather than a static collection of facts.

Temporal studies reinforce this view. Liu et al. (2021), examining RoBERTa during pretraining, found that different kinds of knowledge are acquired at different rates: linguistic knowledge is acquired comparatively quickly and robustly, whereas factual and commonsense knowledge can be slower and more domain-sensitive. This suggests that a model's temporal knowledge boundary is unlikely to be perfectly uniform.

2.2 Reported Versus Effective Knowledge Cutoffs

A model's advertised knowledge cutoff is usually a convenient summary of its training-data boundary. It should not, however, be treated as a precise timestamp at which all knowledge begins or ends.

Cheng et al. (2024) introduced the concept of an **effective cutoff**, defined at the level of resources or topics rather than necessarily at the model level. Their analysis found discrepancies between reported and effective cutoffs, attributing some discrepancies to the temporal structure of Common Crawl data and to deduplication procedures.

This distinction is crucial for scientific literature reviews. Scientific publications have heterogeneous publication dates, indexing delays, versions, preprints, conference proceedings, corrections, retractions, and citation relationships. Consequently, a model may contain knowledge about an older version of a paper while lacking its subsequent correction, or it may have partial information about a research topic while missing important recent developments.

Knowledge cutoff should therefore be modeled not as a single date but as a distribution:

[
$K(m,d,s)$
]

where (m) represents the model, (d) the domain or topic, and (s) the information source. The effective temporal boundary can vary with all three variables.

2.3 Temporal Generalization

Temporal generalization evaluates whether models remain reliable as information changes over time. Chen et al. (2021) introduced a time-sensitive question-answering dataset specifically because conventional QA benchmarks contain relatively few questions whose answers evolve over time. Their experiments showed substantial difficulty in temporal reasoning, with the best evaluated system achieving only 46% accuracy compared with 87% human performance.

Zhu et al. (2024) similarly showed that traditional static benchmarks can fail to capture the changing information environment faced by modern LLMs. Their temporal-generalization framework found performance degradation as the temporal distance from relevant information increased.

More recent research provides an even broader perspective. Li et al. (2026) argue that static pretrained models operating in dynamic environments experience multiple temporal failure modes, including stale world knowledge, inconsistent timelines, incorrect temporal anchoring, and hallucinated future information.

These findings directly apply to literature review. Scientific synthesis requires not merely knowing facts but knowing **when those facts were established, revised, superseded, or contested**.

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3. Knowledge Cutoff Effects on LLM-Generated Literature Reviews

3.1 Loss of Recency and Coverage

The most direct cutoff effect is incomplete coverage. Consider a field in which publications ($P_1,\ldots,P_n$) appear over time. An LLM with cutoff (t_c) can only directly rely on publications available within its effective knowledge boundary. A review generated at ($t_q>t_c$) therefore operates over an incomplete set:

$$
P_{\text{model}} = {p_i : t_i \leq t_c}
$$

rather than the complete set relevant to the review:

$$
P_{\text{current}} = {p_i : t_i \leq t_q}.
$$

The missing set,

$$
P_{\text{missing}} = P_{\text{current}} - P_{\text{model}},
$$

may contain only incremental papers in a mature field, but it can be decisive in rapidly evolving research areas. A new benchmark, model architecture, clinical trial, regulatory decision, or negative replication can substantially alter the interpretation of earlier research.

This creates a distinction between **factual correctness** and **review completeness**. An LLM statement can be factually correct when evaluated against an older paper while still being misleading because newer evidence has changed the field.

3.2 Outdated Consensus

Cutoffs can cause models to overrepresent historically dominant approaches. In rapidly developing areas, older methods may remain statistically prominent in training data even after newer approaches become standard practice.

For example, an AI literature review generated from a corpus ending before a major methodological transition may describe an earlier paradigm as "state of the art." The statement may have been accurate at (t_c), but not at (t_q). The problem is not necessarily hallucination; it is **temporal misrepresentation**.

This distinction matters because conventional factuality benchmarks often ask whether a statement is true, rather than whether it was true *at the appropriate time*. Temporal literature-review evaluation therefore requires at least three dimensions:

- **Truth:** Is the statement supported?
- **Freshness:** Is the evidence sufficiently current?
- **Temporal validity:** Is the statement correctly anchored to the period being discussed?

3.3 Citation Incompleteness and Citation Hallucination

A cutoff can also distort references. If an LLM has incomplete access to the current literature, it may substitute older papers for newer ones, omit relevant papers, or generate plausible but nonexistent references.

Liu et al. (2023) evaluated citation behavior in generative search systems and found that only 51.5% of generated sentences were fully supported by citations, while 74.5% of citations supported their associated statements. These results demonstrate that citation presence should not be confused with citation validity.

Recent scientific-literature experiments provide an even more direct warning. Asai et al. (2026) report that, in their experiments, GPT-4o fabricated citations in 78–90% of cases when asked to cite recent literature across computer science and biomedicine. This finding is particularly relevant to knowledge-cutoff effects because asking for recent literature forces the model beyond what can safely be supplied by its parametric memory.

The implication is that an LLM-generated literature review should be treated as an **evidence-grounded synthesis task**, not as ordinary text generation.

3.4 Temporal Inconsistency

A more subtle problem occurs when a generated review combines evidence from different temporal periods without making the chronology explicit. Suppose a model correctly states that method A was superior in 2021 and method B was superior in 2025. If the review combines both claims without temporal qualification, readers may infer that the two findings are directly comparable.

Temporal inconsistency can occur at several levels:

1. **Event level:** incorrectly ordering scientific developments.
2. **Claim level:** presenting historical findings as current.
3. **Citation level:** citing an old study for a newer claim.
4. **Synthesis level:** combining evidence from incompatible research periods.
5. **Consensus level:** treating an outdated majority view as the current consensus.

Recent temporal research characterizes these problems as part of a broader failure of static models operating in dynamic environments (Li et al., 2026).

3.5 Research-Field Drift

Rapidly evolving fields also experience **concept drift**. Terminology, benchmarks, evaluation standards, and research priorities change. A model trained on earlier literature may understand the words but misunderstand their current technical significance.

For example, the meaning of "state of the art," "efficient," "agentic," "reasoning," or "retrieval" may shift as methods evolve. Thus, the problem is not only missing new papers; the model may also interpret new concepts through an older conceptual framework.

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4. Main Technical Discussion

4.1 Why Larger Context Windows Do Not Eliminate Cutoff Effects

One might assume that increasing an LLM's context window solves knowledge freshness because more documents can be supplied at inference time. However, context capacity and information quality are distinct.

Liu et al. (2024) demonstrated the "lost in the middle" phenomenon: models can perform substantially worse when relevant information is located in the middle of a long context than when it occurs near the beginning or end.

For literature reviews, this creates a practical problem. A system may retrieve hundreds of relevant papers, but the model may fail to use important evidence because:

- relevant evidence is buried among distractors;
- papers repeat similar claims;
- newer evidence conflicts with older evidence;
- the model has limited attention to individual passages;
- context compression removes temporal metadata;
- papers vary greatly in methodological quality.

Thus, simply increasing the number of retrieved documents does not guarantee a better review.

4.2 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) was proposed by Lewis et al. (2020) as a method for combining parametric language-model memory with non-parametric external memory. The original work demonstrated that retrieval could improve factuality and specificity in knowledge-intensive tasks while addressing the difficulty of updating model knowledge.

The basic RAG process can be expressed as:

$$
D_q = R(q, C)
$$

where (q) is the research query, (C) is an external corpus, and (R) is a retrieval function. The LLM then generates:

$$
y = G(q,D_q;\theta).
$$

For literature reviews, (C) may be a scholarly database, institutional repository, preprint server, or specialized scientific corpus.

RAG changes the temporal problem from "Can the model remember recent literature?" to "Can the system retrieve, interpret, and correctly synthesize recent literature?"

This is a major improvement, but it introduces new failure points.

4.3 Retrieval Quality as a Temporal Bottleneck

A literature-review system can only synthesize what it retrieves. If the retrieval index is outdated, the RAG system inherits the same knowledge cutoff.

A current database can still produce poor results if retrieval ranks older papers above newer ones. Therefore, retrieval systems for scientific review should consider temporal relevance explicitly:

$$
S(p,q)=\alpha R_{\text{semantic}}(p,q)
+\beta R_{\text{quality}}(p)
+\gamma R_{\text{temporal}}(p,q)
$$

where (R_{semantic}) measures topical relevance, (R_{quality}) reflects evidence quality, and (R_{temporal}) represents temporal suitability.

The temporal component should not always favor the newest paper. In a historical review, older studies are essential. The appropriate objective is **query-conditioned temporal relevance**, not universal recency.

4.4 Self-Reflective and Corrective Retrieval

Self-RAG extends conventional retrieval by allowing the model to decide when retrieval is needed and to critique retrieved evidence and its own output. Asai et al. (2024) report improvements in factuality and citation accuracy for long-form generation compared with standard approaches.

Corrective RAG (CRAG) addresses another weakness: retrieval itself may be wrong. Yan et al. (2024) proposed a retrieval evaluator that assesses retrieved evidence and triggers corrective retrieval or web search when the initial evidence is inadequate.

For literature reviews, these approaches suggest a useful architecture:

query → retrieval → retrieval evaluation → evidence extraction → synthesis → citation verification → temporal consistency check → final review

This pipeline is more appropriate than single-pass generation because it separates retrieval quality from generation quality.

4.5 Automated Survey Generation

AutoSurvey directly targets automated literature-survey generation. Wang et al. (2024) describe a pipeline involving retrieval and outline generation, subsection drafting, integration, refinement, and evaluation. The work explicitly identifies context limitations and parametric knowledge constraints as challenges to automated survey construction.

The significance of this approach is methodological: a long literature review should not necessarily be generated in one pass. Decomposing the task into smaller components can reduce context pressure and make source attribution more explicit.

However, decomposition does not by itself solve temporal validity. A system can systematically generate an outdated survey if its retrieval corpus is outdated.

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5. Current Approaches and Developments

5.1 Search-Engine-Augmented LLMs

Vu et al. (2024) introduced FreshLLMs and the FreshQA benchmark to evaluate models against dynamically changing world knowledge. Their evaluation involved more than 50,000 human judgments and found that LLMs struggle with rapidly changing information and false-premise questions.

The approach demonstrates the importance of evaluating LLMs using temporally dynamic datasets rather than relying solely on static benchmarks.

For scientific literature reviews, the equivalent is a benchmark in which the same research question is evaluated at multiple points in time, with the expected evidence set changing as publications appear.

5.2 OpenScholar and Scientific Literature Retrieval

A major recent development is OpenScholar, introduced by Asai et al. (2026). OpenScholar combines an open scientific datastore containing approximately 45 million open-access papers with specialized retrieval, reranking, an LLM, and iterative self-feedback. It was evaluated using ScholarQABench, a multidisciplinary benchmark covering computer science, physics, neuroscience, and biomedicine.

The reported results are notable because OpenScholar-8B outperformed GPT-4o and PaperQA2 on a multi-paper synthesis task, while OpenScholar's retrieval-and-feedback pipeline also improved GPT-4o relative to using GPT-4o alone.

This development illustrates a fundamental shift: rather than attempting to make the parametric model itself contain all scientific knowledge, the system separates **reasoning and generation** from **current scientific memory**.

5.3 Temporal Knowledge Graphs

Another approach is to represent evolving knowledge using temporal knowledge graphs. Di Maio et al. (2025) proposed integrating LLMs with dynamic temporal knowledge graphs so that fresh information can be continually stored and updated without relying exclusively on model fine-tuning.

Temporal graphs are attractive for literature reviews because scientific relationships are inherently temporal. A graph can represent:

- paper A precedes paper B;
- method A improves on method B;
- study C replicates study D;
- claim E is contradicted by study F;
- benchmark G supersedes benchmark H.

This allows a review system to represent not only *what* is known but *when* and *how* the knowledge changed.

5.4 Time-Aware Retrieval

Recent research increasingly treats temporal information as an explicit retrieval dimension. Liu, Cao, and Li (2026) found that conventional RAG and several learning-based approaches can struggle with continuous knowledge drift, including temporal inconsistency and catastrophic forgetting. Their Chronos approach organizes retrieved evidence into an Event Evolution Graph to improve temporal consistency.

This is particularly relevant to literature review because scientific progress is cumulative rather than simply additive. A current review needs to understand relationships among old and new evidence, not merely retrieve the latest papers.

5.5 Cutoff Simulation and Temporal Contamination

Gao et al. (2025) studied whether prompting can force LLMs to simulate an earlier knowledge cutoff. Their experiments found that prompting can reduce direct access to post-cutoff information but is less successful when the information is indirectly or causally related to the query.

This result has an important methodological implication: simply instructing an LLM to "ignore papers after 2022" is not sufficient to guarantee temporal isolation.

For research evaluation, this means that benchmark designers must control for contamination rather than assuming that a textual instruction creates a genuine temporal boundary.

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6. Research Challenges and Limitations

6.1 No Universal Definition of "Current"

A literature review can be "current" relative to different objectives. A historical review may intentionally prioritize older evidence, whereas a review of an active AI research area may require papers published within weeks or months.

Therefore:

$$
\text{Freshness} \neq \text{Recency alone}.
$$

Freshness should be defined relative to the research question, field velocity, evidence lifecycle, and intended use.

6.2 Scientific Publication Delays

Publication date is not necessarily equivalent to discovery date. A paper may first appear as a preprint, then undergo peer review, conference publication, journal publication, correction, or retraction. A system using only publication dates may therefore construct an inaccurate temporal sequence.

6.3 Conflicting Evidence

Newer evidence is not automatically better evidence. A recent small study can be less reliable than an older, large-scale investigation. Temporal ranking therefore cannot replace scientific-quality assessment.

An ideal literature-review system should jointly model:

[
\text{Evidence Value} =
f(\text{relevance},\text{quality},\text{recency},\text{replication},\text{citation context}).
]

6.4 Citation Verification

Citation generation remains a significant limitation. The existence of a plausible reference is not sufficient; the reference must actually support the claim. Liu et al. (2023) demonstrated the gap between citation presence and citation support, while OpenScholar's evaluation shows that specialized retrieval and verification can substantially improve citation reliability.

6.5 Long-Context Limitations

Large literature collections can overwhelm the generation stage. The "lost in the middle" finding indicates that simply placing more documents into the context may reduce the reliability of information use.

Consequently, future systems require evidence selection and compression mechanisms that preserve:

- publication dates;
- methodological details;
- sample characteristics;
- outcome measures;
- limitations;
- contradictory findings;
- citation relationships.

6.6 Evaluation Limitations

Current benchmarks often emphasize question answering rather than genuine literature synthesis. OpenScholar's ScholarQABench is an important step because it evaluates long-form, multi-paper scientific synthesis and combines automatic and expert evaluation.

Nevertheless, even such benchmarks cannot fully capture real-world scholarly review, which may involve thousands of papers, multiple databases, changing eligibility criteria, expert judgment, and months of research activity.

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7. Research Gaps and Future Directions

7.1 Dynamic Longitudinal Benchmarks

The most important research gap is the absence of sufficiently comprehensive longitudinal benchmarks.

A future benchmark should freeze the scholarly corpus at multiple dates:

$$
C_{t_1}, C_{t_2}, \ldots, C_{t_n}
$$

and ask the same literature-review question at each point. The benchmark can then evaluate whether the model:

1. identifies newly relevant papers;
2. removes superseded claims;
3. preserves valid historical findings;
4. updates consensus appropriately;
5. maintains citation accuracy;
6. recognizes unresolved disagreements.

Such a benchmark would directly measure **temporal robustness**.

7.2 Temporal Provenance as a First-Class Property

Current citation systems usually answer "Where did this statement come from?" A temporal literature-review system should additionally answer:

- When was the evidence published?
- Was it current at the time of the claim?
- Has it subsequently been challenged?
- Is the cited finding still supported?
- What newer evidence changes the interpretation?

This suggests the concept of **temporal provenance**:

$$
P(c)=\{s,t,v,r\}
$$

where (s) is source identity, (t) is temporal information, (v) is source version, and (r) represents the relationship of later evidence to the claim.

7.3 Evidence Graphs

Rather than representing literature as a flat collection of documents, systems should construct evidence graphs containing relationships such as:

supports → contradicts → extends → replicates → supersedes → corrects

Such structures could help LLMs distinguish an influential historical paper from the currently accepted evidence.

7.4 Adaptive Retrieval Based on Field Velocity

Different scientific fields evolve at different speeds. AI research may require weekly or monthly updating, whereas some areas of mathematics may remain stable for years.

A future retrieval system could estimate field velocity:

$$
V_f = \frac{\text{rate of relevant evidence change}}{\text{unit time}}
$$

and adapt its freshness threshold accordingly.

7.5 Automatic Temporal Consistency Checking

Before producing a final review, the system should perform a dedicated temporal audit. A verifier could detect statements such as:

> "Method X is currently state of the art."

and require evidence dated sufficiently close to the review date.

Similarly, statements about consensus should require multiple temporally appropriate sources rather than a single old citation.

7.6 Human-in-the-Loop Review

The goal should not necessarily be to eliminate human researchers. Instead, LLMs can perform retrieval, clustering, preliminary synthesis, and evidence organization while researchers perform high-level judgment.

A useful division of labor is:

LLM: retrieval → screening → extraction → clustering → preliminary synthesis

Researcher: eligibility judgment → methodological assessment → interpretation → final claims

This model recognizes that scientific review contains epistemic decisions that cannot be reduced to textual similarity.

7.7 Benchmarking Citation Quality Separately from Text Quality

A future evaluation framework should report at least five independent dimensions:

Dimension	Core question
Coverage	Did the system identify important relevant literature?
Freshness	Did it incorporate sufficiently recent evidence?
Citation accuracy	Does each citation support the associated claim?
Temporal consistency	Are claims correctly anchored in time?
Synthesis quality	Does the review explain relationships among studies?

A single overall score can obscure important failure modes. A model could write excellent prose while having poor citation accuracy or weak temporal coverage.

7.8 Continual Updating Without Catastrophic Forgetting

The 2026 literature on continuous knowledge drift suggests that simply updating models is itself difficult. Liu et al. (2026) report that existing learning-based approaches can encounter catastrophic forgetting and temporal inconsistency under continuously changing information.

A promising architecture is therefore a **hybrid temporal memory** in which stable general reasoning remains in model parameters while changing scientific evidence is stored externally. The model would not need to memorize every new paper; it would need reliable mechanisms for retrieving and reasoning over evolving evidence.

8. Conclusion

Knowledge cutoffs represent a fundamental limitation for LLM-generated literature reviews in rapidly evolving scientific fields. The problem is more complex than the absence of post-cutoff information. Research demonstrates that LLMs possess heterogeneous effective temporal boundaries, that temporal reasoning can degrade as information changes, and that static models can generate outdated or temporally inconsistent outputs even when their statements appear fluent and plausible.

For literature review, the consequences are particularly important because a review is fundamentally a representation of the state of knowledge at a particular time. A model operating beyond its effective cutoff can omit new studies, misidentify the current research frontier, overrepresent obsolete methods, reproduce outdated consensus, and generate unreliable or fabricated citations.

Retrieval-augmented approaches provide the most practical current solution because they separate the model's parametric reasoning capabilities from an updateable external evidence store. Foundational RAG, Self-RAG, corrective retrieval, search-engine augmentation, scientific retrieval systems such as OpenScholar, and emerging temporal knowledge-graph approaches demonstrate substantial progress.

However, retrieval alone is insufficient. A reliable literature-review system must know not only which documents are relevant, but also which evidence is current, which findings have been superseded, how contradictory studies should be reconciled, and whether each generated claim is correctly attributed. The emerging literature on temporal knowledge and continuous knowledge drift supports this broader interpretation.

The next generation of LLM-based literature-review systems should therefore treat **time, provenance, and evidence relationships as first-class components of scientific reasoning**. Evaluation should move beyond conventional factuality toward longitudinal measures of coverage, freshness, citation correctness, temporal consistency, and synthesis quality. Dynamic benchmarks that reconstruct the scholarly state of a field at different points in time will be particularly valuable.

Ultimately, LLMs are most promising not as static replacements for scholarly judgment but as continuously connected research assistants whose external evidence, retrieval processes, temporal assumptions, and citations can be inspected and audited. Such systems could substantially reduce the burden of literature synthesis while preserving the epistemic standards required for trustworthy scientific research.

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